

Notes: Coval, Jurek, and Stafford (AER, 2009)
Economic Catastrophe Bonds

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1 Big picture

- **Question.** The paper asks whether structured finance securities with very high credit ratings are actually safe from an asset-pricing perspective. The core issue is whether credit ratings, which summarize expected losses or default probabilities, are enough for pricing securities whose losses occur mainly in bad aggregate states.
- **Main object.** The paper studies collateralized debt obligation (CDO) tranches, especially tranches referencing the Dow Jones CDX North America Investment Grade index. These are claims on a diversified portfolio of corporate credit risks.
- **Central idea.** Pooling and tranching economic assets reallocates losses across states of the world. Senior tranches have low unconditional default probabilities, but when they lose money, they tend to lose in severe recessions or market crashes.
- **Economic catastrophe bond.** A senior tranche is like a bond that pays normally in most states but fails exactly when marginal utility is high. This is why the authors call such securities “economic catastrophe bonds.”
- **Why ratings are insufficient.** A credit rating describes expected payoff information, such as default probability and expected loss. It does not tell investors *when* default happens. A security that defaults in ordinary idiosyncratic states and a security that defaults only in market disasters can have the same rating but should have very different prices.
- **Method.** The authors build an Arrow-Debreu state-contingent pricing framework. They define economic states using the realized market return, extract state prices from S&P 500 index options, and use a modified Merton credit model to map market states into credit losses.
- **Empirical setting.** The empirical sample covers September 2004 to September 2007 and uses CDX index spreads, CDX tranche spreads, corporate bond spread benchmarks, and five-year S&P 500 option quotes.
- **Main result.** Senior and mezzanine CDX tranches offered much too little compensation for their systematic risk. In the baseline model, the 7–10 tranche has an average model spread of 150 bps versus an actual spread of 39 bps, and the 10–15 tranche has an average model spread of 87 bps versus an actual spread of 18 bps.
- **Replicating-risk comparison.** Investors could earn much larger spreads by taking comparable market-risk exposure in index option markets, for example by writing out-of-the-money put spreads.
- **Economic takeaway.** Structured finance can manufacture securities that look safe by rating but are extremely exposed to aggregate bad states. The key asset-pricing lesson is that expected loss is not enough; the state-contingent distribution of losses matters.

2 Incremental contribution of the paper

- **From default probabilities to state-contingent pricing.** Earlier credit analysis often emphasized expected loss, credit ratings, or reduced-form default intensities. This paper stresses that the state in which default occurs is central for pricing.

- **A formal theory of structured finance as risk concentration.** The paper shows analytically that, holding the unconditional default probability fixed, increasing diversification in the underlying pool makes the senior tranche payoffs more concentrated away from bad states. This lowers the correct price and raises the required spread.
- **Cheapest-to-supply logic.** The paper proves that the cheapest security to supply for a target expected loss is a security that fails exactly in the worst economic states. Highly diversified senior tranches approximate this object.
- **Operational Arrow-Debreu implementation.** The paper does not stop at theory. It extracts state prices from index options and uses them to price corporate credit and tranche payoffs in a unified framework.
- **Bridge across markets.** The empirical design links three markets: the S&P 500 option market, the corporate credit/CDX market, and the CDX tranche market. The authors first calibrate to the CDX index, then ask whether tranche prices are consistent with the same state prices.
- **Evidence on structured credit mispricing before the crisis.** Using 2004–2007 data, the paper finds that senior tranche spreads were close to rating-matched corporate bond spreads even though the tranche losses were much more concentrated in bad market states.
- **Replication benchmark.** The paper gives a transparent comparison: put-spread positions on the market index can replicate much of the senior-tranche market exposure and offer much larger compensation.
- **Reinterpretation of the equity tranche.** The paper argues that the first-loss equity tranche may be less toxic from a systematic-risk perspective than commonly believed, because it bears many idiosyncratic first losses. In the model, equity tranches appear underpriced while nonequity tranches appear overpriced.
- **Ratings critique.** The paper provides a clean asset-pricing critique of ratings-based investing: ratings can rank expected losses but cannot determine discount rates because they ignore state prices.

3 Data and measurement

3.1 CDX index and CDX tranche data

- The main credit-market object is the Dow Jones CDX North America Investment Grade index, a portfolio of 125 liquid five-year credit default swap contracts on investment-grade US firms.
- The sample runs from September 2004 to September 2007.
- The authors splice the first six months of CDX Series 3 through Series 8 to create a continuous on-the-run CDX series.
- Tranches are defined by loss attachment points: 0–3, 3–7, 7–10, 10–15, and 15–30 percent. The model also reports a 30–100 super-senior tranche in some exercises.
- A tranche pays fully if portfolio losses remain below its lower attachment point, pays zero if losses exceed its upper attachment point, and pays linearly between those points.

- Tranche prices are quoted as running spreads paid by the protection buyer. The 0–3 equity tranche is normally quoted with an upfront payment plus a fixed running spread, so the authors convert it into an equivalent running spread.

Interpretation: The CDX is a clean setting because it is standardized, liquid, and directly linked to market-observed tranche prices. The tranches provide a natural laboratory for comparing rating-like expected losses with state-contingent risk exposures.

3.2 Corporate bond and rating-matched benchmarks

- The authors compare CDX and tranche spreads with five-year credit default swap spreads implied by corporate bond spreads for AA, A, BBB, BB, and B-rated bonds.
- These rating-group spreads help reveal whether the market prices tranches like rating-matched single-name corporate credit.
- The central empirical pattern is that nonequity CDX tranches trade at spreads similar to rating-matched corporate bonds, even though their losses are more systematic.

Interpretation: Rating-matched corporate bonds are the natural benchmark used by ratings-focused investors, but they are not the correct risk-matched benchmark for senior tranches.

3.3 S&P 500 option data and state prices

- The pricing model requires state prices over five-year market-return states.
- The authors use daily over-the-counter quotes on five-year S&P 500 options from Citigroup.
- Each daily cross section contains 13 standardized moneyness levels from 0.70 to 1.30 in increments of 0.05.
- They fit a smooth implied volatility function, with the preferred specification based on a hyperbolic tangent function.
- They then extract Arrow-Debreu state prices using the Breeden-Litzenberger idea that the second derivative of option prices with respect to strike identifies state prices.

Interpretation: The option market provides a forward-looking price of bad aggregate states. This is exactly the information missing from credit ratings.

3.4 Model calibration inputs

- The CDX portfolio is treated as homogeneous: 125 bonds issued by identical representative firms.
- The representative firm is described by asset beta, debt-to-asset ratio, and idiosyncratic asset volatility.
- The calibration pins down these parameters by matching the actual CDX spread, an equity beta of one, and an average pairwise equity correlation of 0.2.

- Recovery is handled in two ways: a Merton-model recovery value and a state-independent 40 percent recovery assumption.
- The implied volatility function is also handled in two ways: the baseline hyperbolic tangent function and a constrained version that limits extrapolated tail volatility.

Interpretation: The four main model variants are designed to show that the results do not depend on a single recovery-rate or option-tail extrapolation assumption.

4 Models and empirical specifications

4.1 State-contingent pricing idea

The paper starts from Arrow-Debreu pricing. Let s index economic states from bad states to good states. Let $q(s)$ be the state price and $\pi(s)$ be the physical probability of state s . A risky bond with state-dependent default probability $p_D(s)$ and zero recovery has price

$$B_\tau = \int (1 - p_D(s))q(s)ds.$$

- A security is expensive if it pays in bad states, because dollars are valuable in those states.
- A security is cheap if it fails in bad states, because it removes payoffs exactly when investors value payoffs most.
- Credit ratings summarize unconditional default probability or expected loss, not the state-price-weighted value of payoffs.

Interpretation: Pricing requires both expected cash flows and discount rates. Ratings contain information about the first object but not the second.

4.2 Pooling, tranching, and digital tranches

Consider a pool of N homogeneous risky bonds. A simple digital tranche pays one if the portfolio loss is below an attachment point X and zero otherwise. Conditional on state s , portfolio defaults are approximately normal, so the tranche default probability is

$$p_D^X(s) = 1 - \Phi \left(\sqrt{N} \frac{X - p_D(s)}{\sqrt{p_D(s)(1 - p_D(s))}} \right).$$

The unconditional default probability is

$$p_D^X = \int p_D^X(s)\pi(s)ds.$$

- Increasing N diversifies away idiosyncratic default risk.
- For a senior tranche, losses then become concentrated in states where the common economic factor is very bad.

- Holding the unconditional default probability fixed, more diversification means the tranche payoff becomes more disaster-contingent.

Interpretation: Diversification does not eliminate systematic risk. In a tranced structure, it can make systematic risk more concentrated in senior claims.

4.3 Three theoretical propositions

- **Proposition 1.** Holding the unconditional default probability fixed, the value of a digital tranche declines as the number of securities in the underlying pool increases.
- **Proposition 2.** The cheapest security to supply with expected payoff $1 - p^*$ pays zero in the worst p^* measure of states and one elsewhere.
- **Proposition 3.** A tranche written on an asymptotically diversified pool converges to this cheapest-to-supply security.

Interpretation: These propositions show why structured finance can create high-rated securities that are very cheap to produce but should require high risk premia. The trick is not lowering expected loss alone; it is concentrating loss in bad states.

4.4 Merton credit model integrated with a market factor

The empirical model modifies Merton's structural credit model by making firm asset values depend on a common market factor. For firm i ,

$$\ln A_{i,T} - \ln A_{i,t} = \left(r_f + \lambda_A - \frac{\sigma_A^2}{2} \right) \tau + \beta_A \sigma_m \sqrt{\tau} Z_m + \sigma_\epsilon \sqrt{\tau} Z_{i,\epsilon}.$$

The market return is

$$\ln M_T - \ln M_t = \left(r_f - \delta_m + \lambda_m - \frac{\sigma_m^2}{2} \right) \tau + \sigma_m \sqrt{\tau} Z_m.$$

Conditional on market log-moneyness m_T , terminal assets are

$$A_{i,T}(m_T) = A_{i,t} \exp \left(r_f \tau + \beta_A m_T + \sigma_\epsilon \sqrt{\tau} Z_{i,\epsilon} \right).$$

Default occurs when $A_{i,T} < D$, so the conditional default probability is

$$p_D(m_T) = \Phi[-\eta(m_T)], \quad \eta(m_T) = \frac{\ln(A_t/D) + r_f \tau + \beta_A m_T}{\sigma_\epsilon \sqrt{\tau}}.$$

- Market states affect default probability through m_T .
- Bad market states raise conditional default probabilities for many firms at the same time.
- After conditioning on the market state, idiosyncratic shocks determine which firms default within that state.

Interpretation: This setup maps aggregate market outcomes into portfolio credit losses, which is what is needed to price tranche payoffs with option-implied state prices.

4.5 Extracting state prices from options

The state space is defined by market moneyiness,

$$m_T = \ln \left(\frac{M_{t+T}}{FUT_t} \right).$$

In the Breeden-Litzenberger framework,

$$q(m_T) = \frac{\partial^2 C(K, \tau)}{\partial K^2} \Big|_{K=xFUT},$$

where $C(K, \tau)$ is the call price and $x = \exp(m_T)$ is moneyiness. Because index options have a volatility smile, the authors fit an implied volatility function and adjust the state-price formula for the slope and curvature of the smile.

- Deep out-of-the-money put options reveal how expensive market-crash states are.
- Those are precisely the states in which senior tranches lose money.
- The model therefore prices tranche losses using market prices of aggregate disaster risk.

Interpretation: The option market provides the price of insurance against aggregate crashes. Senior CDO tranches are short that insurance.

4.6 Tranche payoff and pricing

For a tranche with lower attachment point X and upper attachment point Y , the normalized payoff is

$$P_T^{[X,Y]}(m_T) = 1 - \frac{1}{Y-X} [\max\{(1-X) - P_T(m_T), 0\} - \max\{(1-Y) - P_T(m_T), 0\}].$$

The model simulates conditional CDX portfolio payoffs, applies tranche contract terms, computes mean state-contingent tranche payoffs, and prices them using option-implied state prices.

- The CDX index is first calibrated so that the model matches the actual CDX spread.
- Then the same model and same state prices are used to price tranches.
- The comparison between model tranche spreads and actual tranche spreads is the main mispricing test.

Interpretation: The empirical design gives the tranches the benefit of a model already calibrated to the underlying CDX. The question is whether tranche prices are consistent with the risks of the underlying credit index.

4.7 Put-spread approximation and credit risk ratio

The authors also approximate tranche exposure with market-index put spreads. A put-spread portfolio is easier to interpret because it uses traded index options instead of a continuum of Arrow-Debreu securities.

The paper defines the credit risk ratio

$$\varphi = \frac{\text{Yield Spread}}{\text{Loss Rate}},$$

where the loss rate is the annualized spread needed to compensate only for expected loss, without a risk premium.

- A high φ means that a security requires a large risk premium per unit of expected loss.
- High-rated single-name corporate bonds have higher φ than low-rated bonds because their rare defaults tend to happen in worse states.
- Senior CDX tranches should have extremely high φ because their losses are even more concentrated in bad states.

Interpretation: The credit risk ratio is a compact way to show how much state-contingent risk is hidden behind a given expected loss.

5 Identification and empirical design

5.1 What the design identifies

- The paper identifies whether observed CDX tranche prices are consistent with a state-contingent pricing model that also fits the CDX index and uses option-implied prices of aggregate market states.
- It is a relative pricing exercise across markets, not a randomized causal design.
- The central comparison is between actual tranche spreads and spreads implied by risk-matched option-market payoffs.
- The paper’s strongest claim is not that every assumption is exactly true, but that rating-matched corporate bonds are the wrong benchmark for senior tranches.

Interpretation: The evidence is best read as an asset-pricing diagnosis: senior tranches were priced as if their rating summarized their risk, even though their losses were concentrated in bad market states.

5.2 Main identifying assumptions

- **Market return as state space.** The S&P 500 market return is assumed to capture the relevant aggregate economic state for pricing.
- **Option-implied state prices.** Five-year index options are assumed to provide useful prices for the aggregate states relevant to CDX tranche losses.
- **Structural credit mapping.** The Merton-CAPM model is assumed to map market states into firm defaults and recoveries sufficiently well.
- **Homogeneous CDX portfolio.** The 125-name CDX is modeled using a representative firm.

- **Credit-market calibration.** The CDX spread is assumed to reflect credit-risk compensation rather than primarily taxes, liquidity, or convenience effects.
- **Market integration.** The option market and corporate credit market are assumed to be sufficiently integrated that state prices from options can be used to price credit risks.

Interpretation: These assumptions are strong, but the paper checks them with multiple robustness exercises and with time-series spread dynamics.

5.3 Robustness and alternative explanations

- The authors test both Merton recovery and state-independent 40 percent recovery.
- They test both an unconstrained implied volatility smile and a constrained smile that prevents very high extrapolated crash-state volatility.
- They show that the model’s predicted CDX spread changes explain a meaningful part of actual CDX spread changes.
- They examine a severe robustness exercise that sets state prices below market moneyness 0.4 to zero and redistributes the mass. Even then, the main overpricing result for 3–7, 7–10, and 10–15 tranches remains.
- They discuss whether the apparent mispricing could instead be in the option market or the CDX market. Their conclusion is that the model can reconcile options with the CDX, but not options, the CDX, and the tranches simultaneously.

Interpretation: The strongest robustness point is that the results are not driven only by extreme states outside historical experience. The senior-tranche undercompensation survives conservative tail assumptions.

5.4 Limitations of the design

- Tests of market efficiency are joint tests of efficiency and the pricing model.
- The homogeneity assumption simplifies the CDX portfolio and may miss industry or firm-level heterogeneity.
- Option tail extrapolation matters conceptually, even though the paper shows robustness to constrained tails.
- Liquidity, capital constraints, funding costs, and institutional mandates can affect observed prices.
- The sample ends in September 2007, before the full crisis repricing of structured credit.

Interpretation: The paper does not prove a frictionless arbitrage in every detail. It shows that the compensation offered by senior structured-credit tranches was far below what investors could obtain for bearing comparable aggregate-market risk elsewhere.

6 Tables and figures

6.1 Figures 1 and 2: Pooling, tranching, and cheapest-to-supply payoffs

Read:

- Figure 1 shows the payoff of a digital tranche as the number of assets in the collateral pool increases.
- With a small pool, default risk is more spread across states. With a large pool, idiosyncratic risk is diversified away and the payoff becomes steep: the tranche pays in good states and fails in bad states.
- Figure 2 compares a bond portfolio and a CDO tranche with the same expected loss.
- The tranche shifts payoff away from recession states and toward boom states relative to the bond portfolio.

Economic message: The theoretical mechanism is visible in these figures. Pooling and tranching do not merely reduce default probability; they reshape when losses happen. Senior tranches with good ratings can be bad-state claims.

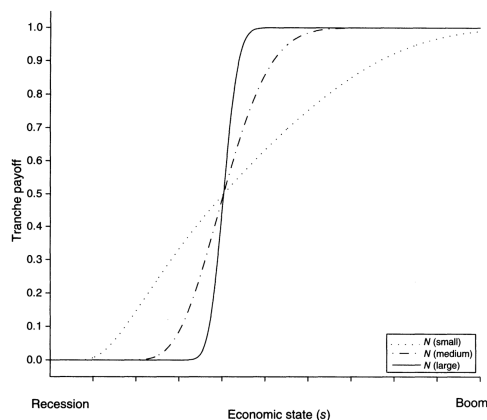


FIGURE 1. THE EFFECT OF COLLATERAL POOL DIVERSIFICATION ON THE STATE-CONTINGENT PAYOFFS OF A CDO TRANCHE

Notes: This figure displays the state-contingent payoff for a CDO tranche that pays one if the losses on the underlying portfolio are less than a prespecified value, X , and pays zero otherwise (a digital tranche). The underlying portfolio is comprised of N identical assets, where N is either 10, 100, or 1,000.

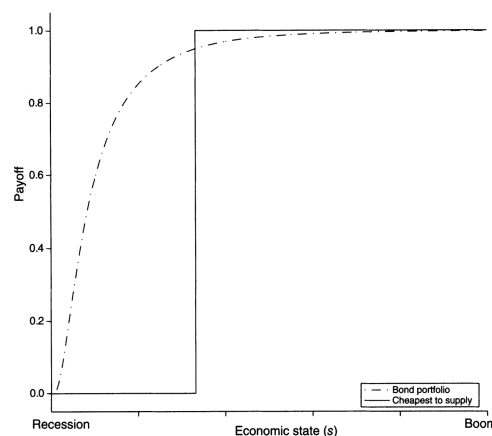


FIGURE 2. STATE-CONTINGENT PAYOFFS OF A BOND PORTFOLIO AND A CDO TRANCHE WITH THE SAME EXPECTED LOSS

Note: This figure displays the state-contingent payoffs for a CDO collateral pool (bond portfolio) and an associated digital CDO tranche that is constructed to have the same expected loss as the underlying collateral.

- (a) Figure 1: collateral pool diversification and tranche payoffs. (b) Figure 2: bond portfolio versus cheapest-to-supply payoff.

6.2 Table 1: Summary statistics for the US credit market

Read:

- Panel A reports average levels and standard deviations for implied volatility, corporate bond rating spreads, the CDX, and CDX tranches.
- The average CDX spread is 45.9 bps, close to the BBB spread of 47.9 bps.

- The 0–3 equity tranche has a much higher average spread than senior tranches, reflecting first-loss exposure.
- Panels B and C show correlations in levels and changes. CDX tranche spreads are highly correlated with the CDX and with each other.
- Changes in five-year implied volatility are positively correlated with changes in credit spreads, consistent with volatility being a key credit-risk state variable.

Economic message: The summary statistics show that CDX tranche prices move with broad credit and volatility conditions. The average spread levels also reveal the puzzle: senior tranches have low quoted spreads despite exposure to aggregate bad states.

TABLE 1—SUMMARY STATISTICS FOR US CREDIT MARKET (9/2004–9/2007)

	σ_{5y}	AA	BBB	CDX	[0–3]	[3–7]	[7–10]	[10–15]	[15–30]
<i>Panel A. Time series means and standard deviations of daily series</i>									
Mean	18.20%	15.60	47.90	45.90	34.00%	138.00	39.10	18.20	8.60
Standard	1.40%	5.00	7.10	10.10	8.50%	61.40	25.10	11.00	5.30
<i>Panel B. Correlations between weekly series</i>									
σ_{5y}	1.00								
AA	0.59	1.00							
BBB	0.61	0.77	1.00						
CDX	0.20	0.50	0.51	1.00					
[0–3]	0.19	0.47	0.49	0.93	1.00				
[3–7]	−0.03	0.26	0.32	0.80	0.66	1.00			
[7–10]	−0.18	0.18	0.23	0.70	0.55	0.95	1.00		
[10–15]	0.00	0.33	0.37	0.87	0.73	0.97	0.94	1.00	
[15–30]	0.17	0.47	0.49	0.95	0.86	0.81	0.73	0.89	1.00
<i>Panel C. Correlations between changes in weekly series</i>									
σ_{5y}	1.00								
AA	0.22	1.00							
BBB	0.13	0.67	1.00						
CDX	0.45	0.28	0.37	1.00					
[0–3]	0.38	0.14	0.25	0.69	1.00				
[3–7]	0.37	0.10	0.16	0.67	0.87	1.00			
[7–10]	0.32	0.18	0.21	0.69	0.77	0.91	1.00		
[10–15]	0.37	0.23	0.27	0.76	0.76	0.88	0.94	1.00	
[15–30]	0.37	0.26	0.26	0.75	0.73	0.77	0.79	0.86	1.00

Notes: This table reports summary statistics for various credit market securities. Rating group indices represent the five-year credit default swap spreads implied by spreads of corporate bonds with the associated credit rating. The CDX series is the Dow Jones CDX North America Investment Grade index of five-year credit default swaps. Spreads of tranches referencing the CDX are denoted by their lower, X, and upper, Y, percentage loss attachment points, [X–Y]. All credit spreads, except for the [0–3] tranche, are reported in basis points (1bps = 0.01 percent). Five-year at-the-money implied volatility from S&P 500 index options is denoted as σ_{5y} .

A Summary Statistics

Table 1: Summary statistics for US credit market securities.

6.3 Figure 3: Implied volatility functions and state prices

Read:

- The top panel plots fitted five-year S&P 500 implied volatility functions for selected CDX initiation dates.
- The downward-sloping implied volatility smile means low market states are expensive to insure against.
- The bottom panel plots the corresponding option-implied state prices.
- The state-price density puts economically important value on bad market states, which are the states in which senior tranches lose money.

Economic message: The option market prices crash states as valuable. If senior tranches fail in those states, then their spreads must compensate investors for selling expensive disaster insurance.

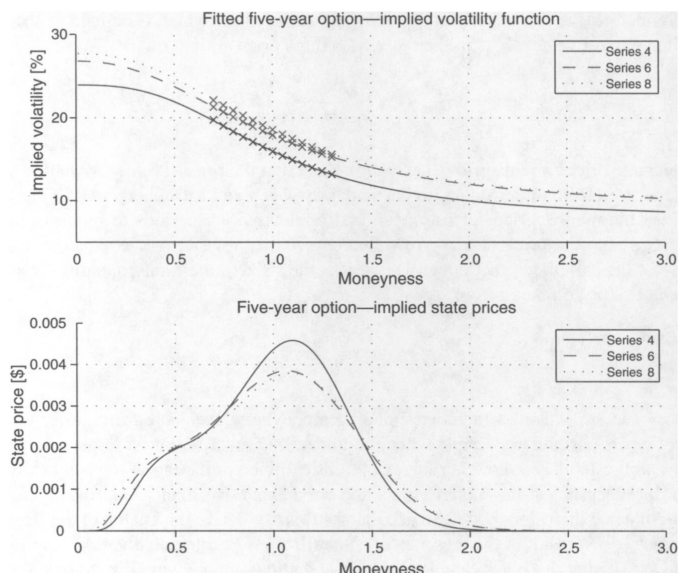


FIGURE 3. CALIBRATED FIVE-YEAR IMPLIED VOLATILITY FUNCTIONS AND STATE PRICES AS OF SELECTED CDX INITIATION DATES

s: The top panel shows the parametric hyperbolic tangent implied volatility function fitted using prices of five-year CDX index options. The values of the actual implied volatilities are denoted by "x." The observed option prices are from a daily cross section of 13 securities with strike prices that have been normalized by the five-year S&P 500 futures price (moneyness), ranging from 0.7 to 1.3, at 0.05 increments. The bottom panel displays the corresponding state prices calculated using the technique of Breeden and Litzenberger (1978), adjusted to account for the implied volatility skew.

Figure 3: Calibrated five-year implied volatility functions and option-implied state prices.

6.4 Tables 2 and 3: Calibration and CDX spread dynamics

Read:

- Table 2 reports calibrated model parameters under four model variants.
- The parameters are similar across variants: asset beta is around 0.71–0.74, debt-to-asset ratio around 0.34–0.38, and idiosyncratic asset volatility around 0.26–0.27.
- Model-implied five-year default probabilities are around 4.1–4.2 percent under the pricing measure, and model recovery rates are around 40–41 percent.
- Table 3 tests whether the model explains CDX spread changes.
- The model-predicted spread change is strongly significant and explains about one-third of weekly CDX spread changes in the simple regressions.

Economic message: The calibration is not forcing unreasonable parameters. The model also captures meaningful CDX spread dynamics, which supports using the calibrated framework to price tranches.

TABLE 2—CDX CALIBRATION RESULTS (9/2004–9/2007)

Model	Asset beta	Debt-to-asset ratio	Idiosyncratic asset volatility	Model 5-yr default probability	Model recovery rate
Merton recovery (1)	0.7359 (0.0196)	0.3435 (0.0249)	0.2688 (0.0140)	0.0410 (0.0029)	0.4103 (0.0042)
State-independent recovery (2)	0.7317 (0.0186)	0.3494 (0.0236)	0.2672 (0.0136)	0.0423 (0.0032)	0.4000 (0.0000)
Merton recovery with constrained implied volatility (3)	0.7150 (0.0197)	0.3725 (0.0251)	0.2612 (0.0139)	0.0417 (0.0030)	0.4110 (0.0043)
State-independent recovery with constrained implied volatility (4)	0.7133 (0.0187)	0.3750 (0.0237)	0.2605 (0.0135)	0.0423 (0.0032)	0.4000 (0.0000)

Notes: This table reports the time series summary statistics of the daily calibrated model parameters, and the model-implied default probabilities and recovery values for four variants of the bond pricing model, calibrated to match the actual CDX spread on each day. The CDX series is the Dow Jones CDX North America Investment Grade index of five-year credit default swaps. All four models assume a hyperbolic tangent implied volatility function. Models 1 and 3 are based on the Merton (1974) credit model recovery value. Models 2 and 4 assume state-independent recovery values of 40 percent. Models 3 and 4 constrain the daily maximum implied volatility to be equal to the maximum observed implied volatility from the daily cross section of five-year index option prices. The three model parameters are referred to as the asset beta (β_A), debt-to-asset ratio (D/A), and idiosyncratic asset volatility (σ_ε). The model-implied five-year default probability and recovery rate are calculated under the pricing measure. The standard deviations of weekly changes in the parameters are reported in parentheses.

(a) Table 2: CDX calibration results.

TABLE 3—REGRESSIONS EXPLAINING THE TIME SERIES OF CHANGES IN CDX YIELD SPREADS (9/2004–9/2007)

Intercept	$E[\Delta y_{1t}]$	$E[\Delta y_{2t}]$	$E[\Delta y_{3t}]$	$E[\Delta y_{4t}]$	Δr_f	$\Delta S\&P$	$\Delta \sigma_{5y}$	ΔSkew	R^2 [N]
0.0000 (1.47)					−0.0687 (−3.16)	−0.0096 (−5.16)	0.0188 (3.50)	0.0089 (3.00)	0.4138 [146]
0.0000 (0.78)	0.3787 (8.70)								0.3402 [146]
0.0000 (1.37)	0.6100 (2.87)				−0.0088 (−0.30)	0.0022 (0.49)	−0.0152 (−1.17)	0.0169 (4.21)	0.4423 [146]
0.0000 (0.85)		0.4050 (8.89)							0.3497 [146]
0.0000 (1.37)		0.7217 (3.04)			−0.0002 (−0.01)	0.0043 (0.87)	−0.0179 (−1.36)	0.0172 (4.33)	0.4461 [146]
0.0000 (0.96)			0.4002 (8.70)						0.3400 [146]
0.0000 (1.45)			0.5609 (2.44)		−0.0107 (−0.33)	0.0021 (0.41)	−0.0055 (−0.49)	0.0165 (3.86)	0.4336 [146]
0.0000 (1.01)				0.4200 (8.85)					0.3479 [146]
0.0000 (1.45)				0.6381 (2.56)	−0.0046 (−0.14)	0.0036 (0.66)	−0.0071 (−0.62)	0.0166 (3.97)	0.4360 [146]

Notes: The dependent variable is the weekly change in the CDX spread. The CDX series is the Dow Jones CDX North America Investment Grade index of five-year credit default swaps. The model predicted change in the CDX spread, $E[\Delta y_{it}]$, is calculated as the difference between the model yield at time $t + 1$, using parameters calibrated at time t , and the actual yield at time t , using one of four models. All four models assume a hyperbolic tangent implied volatility function. Models 1 and 3 are based on the Merton (1974) credit model recovery value. Models 2 and 4 assume state-independent recovery values of 40 percent. Models 3 and 4 constrain the daily maximum implied volatility to be equal to the maximum observed implied volatility from the daily cross section of five-year index option prices. The five-year swap rate is denoted r_f . S&P refers to the log level of the S&P 500 index. Five-year at-the-money implied volatility from S&P 500 index options is denoted as $\Delta \sigma_{5y}$. Skew is the linear slope between the implied volatility of five-year 30 percent out-of-the-money options and the implied volatility of five-year at-the-money options. Weekly changes are denoted by Δ . The adjusted R -square is denoted R^2 . t -statistics are in parentheses, and the number of observations is in square brackets.

(b) Table 3: regressions explaining CDX spread changes.

6.5 Figure 4: State-contingent payoffs for the CDX and the 7–10 tranche

Read:

- The top panels show CDX portfolio payoffs as a function of market moneyiness under two recovery assumptions.
- The bottom panels show the 7–10 tranche payoff.
- The 7–10 payoff has a cliff-like shape: it is almost safe in normal states but deteriorates quickly when market moneyiness falls into bad states.
- The comparison between Merton recovery and state-independent recovery shows that the payoff shape is not purely a recovery-rate artifact.

Economic message: The figure makes the economic-catastrophe-bond nature of the tranche visible. The tranche is designed to pay unless the aggregate market state is bad enough, so its risk is highly systematic.

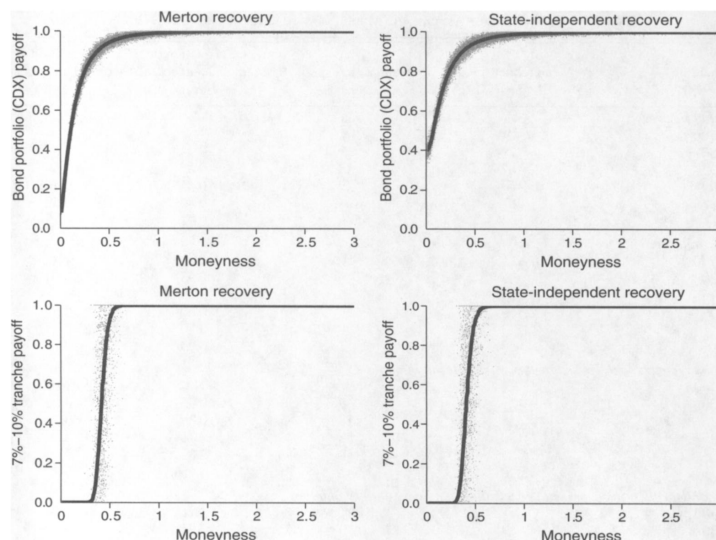


FIGURE 4. SIMULATED STATE-CONTINGENT PAYOFFS FOR THE CDX AND THE 7 PERCENT–10 PERCENT CDX TRANCHE

Notes: The leftmost panels display the CDX and 7 percent–10 percent CDX tranche payoffs using the baseline model specification, which relies on the Merton (1974) credit model recovery assumption. The panels depict the mean state-contingent payoffs (line) and a selection of simulated outcomes (dots). The rightmost panels display these same payoffs calculated with the assumption that recovery rates are state-independent, drawn from a beta distribution with a mean of 0.4 and a standard deviation of 0.25.

Figure 4: Simulated state-contingent CDX and 7–10 tranche payoffs.

6.6 Tables 4 and 5: Actual tranche spreads versus risk-matched alternatives

Read:

- Table 4 compares actual tranche spreads with model-implied spreads.
- For nonequity tranches, model spreads are far above actual spreads. Under Merton recovery, 7–10 is 150 bps model versus 39 bps actual, and 10–15 is 87 bps model versus 18 bps actual.
- The 0–3 equity tranche goes in the opposite direction: actual spreads exceed model spreads.
- Table 5 replaces the full Arrow-Debreu replication with a simpler put-spread approximation.
- Put-spread approximations still imply spreads far above actual spreads for nonequity tranches.

Economic message: These are the central pricing results. Senior and mezzanine CDX tranches appear overpriced because they offer too little yield for the market crash risk they embed. The result survives a simpler, tradable put-spread benchmark.

TABLE 4—COMPARISON OF ACTUAL AND MODEL TRANCHE SPREADS (9/2004–9/2007)

Tranche (percent)	Mean model spread [bps]	Mean actual spread [bps]	Correlation of model and actual (levels)	Correlation of model and actual (changes)
<i>Merton recovery</i>				
30–100	1	NA	NA	NA
15–30	28	9	0.94	0.68
10–15	87	18	0.85	0.66
7–10	150	39	0.67	0.58
3–7	267	138	0.80	0.58
0–3	914	1,508	0.92	0.75
<i>State-independent recovery</i>				
30–100	1	NA	NA	NA
15–30	25	9	0.95	0.70
10–15	85	18	0.85	0.67
7–10	150	39	0.67	0.57
3–7	273	138	0.80	0.58
0–3	953	1,517	0.93	0.75
<i>Merton recovery with constrained implied volatility function</i>				
30–100	<1	NA	NA	NA
15–30	11	9	0.95	0.60
10–15	56	18	0.86	0.69
7–10	130	39	0.71	0.66
3–7	323	138	0.82	0.66
0–3	1,098	1,548	0.95	0.75
<i>State-independent recovery with constrained implied volatility function</i>				
30–100	<1	NA	NA	NA
15–30	9	9	0.95	0.62
10–15	52	18	0.87	0.69
7–10	127	39	0.71	0.65
3–7	327	138	0.82	0.65
0–3	1,124	1,554	0.95	0.75

Notes: Tranches are denoted by their lower and upper percentage loss attachment points. On each day, after the bond pricing model has been calibrated to match the actual CDX spread, model tranche payoffs are obtained by simulating the state-contingent CDX payoffs using the calibrated model parameters and applying the tranche contract terms. Daily model tranche spreads are then computed by valuing the mean simulated state-contingent tranche payoffs using option-implied state prices. The actual CDX tranche spreads correspond to various tranches referencing the Dow Jones CDX North America Investment Grade index of five-year credit default swaps. Correlations of model and actual spreads in levels and changes are computed using weekly time series.

TABLE 5—COMPARISON OF ACTUAL AND PUT SPREAD APPROXIMATED TRANCHE SPREADS (9/2004–9/2007)

Tranche (percent)	Mean spread from put spread approximation [bps]	Mean actual spread [bps]	Correlation of model and put spread (levels)	Correlation of model and put spread (changes)
<i>Merton recovery</i>				
30–100	2	NA	NA	NA
15–30	29	9	0.59	0.20
10–15	82	18	0.45	0.19
7–10	139	39	0.30	0.12
3–7	263	138	0.42	0.09
0–3	837	1,508	0.55	0.34
<i>State-independent recovery</i>				
30–100	1	NA	NA	NA
15–30	26	9	0.59	0.20
10–15	80	18	0.45	0.19
7–10	139	39	0.30	0.12
3–7	269	138	0.42	0.09
0–3	872	1,517	0.54	0.35
<i>Merton recovery with constrained implied volatility function</i>				
30–100	<1	NA	NA	NA
15–30	11	9	0.64	0.15
10–15	51	18	0.46	0.15
7–10	117	39	0.32	0.11
3–7	319	138	0.40	0.12
0–3	1,008	1,548	0.51	0.35
<i>State-independent recovery with constrained implied volatility function</i>				
30–100	<1	NA	NA	NA
15–30	9	9	0.64	0.15
10–15	48	18	0.47	0.15
7–10	115	39	0.33	0.11
3–7	323	138	0.40	0.12
0–3	1,031	1,554	0.49	0.35

Notes: Tranches are denoted by their lower and upper percentage loss attachment points. The model-implied state-contingent tranche payoffs are approximated using a put spread portfolio comprised of a riskless bond and two put options on the S&P 500 index. The spread of the approximating portfolio is obtained on each day by valuing its state-contingent payoffs using option-implied state prices. The composition of the approximating put spread portfolio is readjusted once every six months on the CDX initiation date. The actual CDX tranche spreads correspond to various tranches referencing the Dow Jones CDX North America Investment Grade index of five-year credit default swaps. Correlations of model and actual spreads in levels and changes are computed using weekly time series.

- (a) Table 4: actual versus model tranche spreads. (b) Table 5: actual spreads versus put-spread approximations.

6.7 Figures 5 and 7: Time-series pricing gaps and extreme-state robustness

Read:

- Figure 5 plots model and actual tranche spreads through time.
- For nonequity tranches, model spreads are persistently above actual spreads across the sample.
- The equity tranche displays the opposite pattern: actual spreads are generally above model spreads.
- Figure 7 repeats the exercise after ruling out extreme market states below moneyness 0.4 by assigning those states zero state price.
- Even with this conservative adjustment, the 3–7, 7–10, and 10–15 tranches remain overpriced relative to the model.

Economic message: The pricing gap is persistent, not just a single-date anomaly. It is also not driven only by far-tail disaster states outside the option data. The core undercompensation appears in economically relevant adverse states.

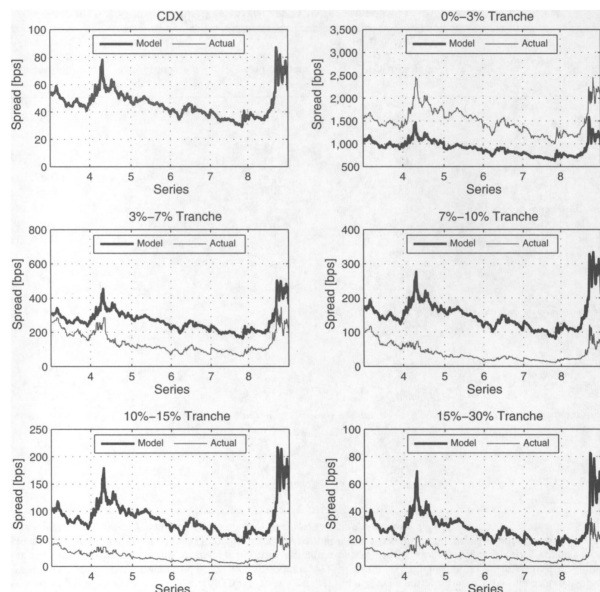


FIGURE 5. TIME SERIES COMPARISON OF MODEL AND ACTUAL TRANCHE SPREADS (9/2004–9/2007)

Notes: On each day, after the bond pricing model has been calibrated to match the actual CDX spread, model tranche payoffs are obtained by simulating the state-contingent CDX payoffs using the calibrated model parameters and applying the tranche contract terms. Daily model tranche spreads are then computed by valuing the mean simulated state-contingent tranche payoffs using option-implied state prices. The actual CDX tranche spreads correspond to various tranches referencing the Dow Jones CDX North America Investment Grade index of five-year credit default swaps.

(a) Figure 5: model versus actual tranche spreads.

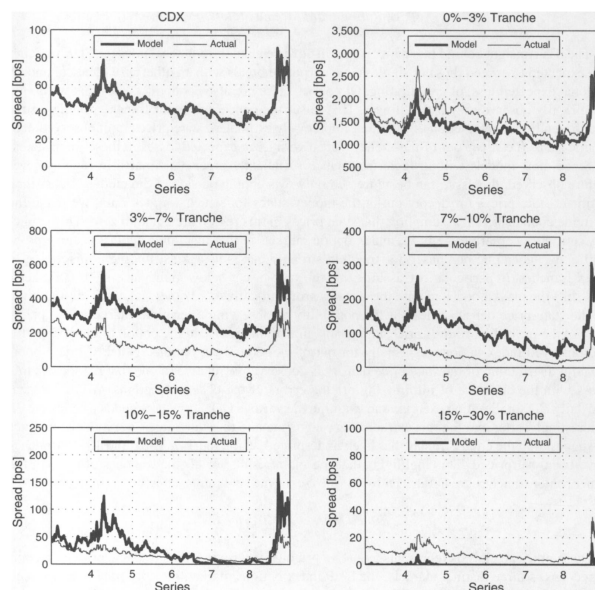


FIGURE 7. ROBUSTNESS OF MODEL RESULTS TO RULING OUT EXTREME MARKET STATES (9/2004–9/2007)

Notes: This figure compares the model and actual tranche spreads from a calibration in which the state prices of states with a moneyness below 0.4 have been set to zero, and the tail weight reallocated proportionally among the remaining states. On each day, after the bond pricing model has been calibrated to match the actual CDX spread, model tranche payoffs are obtained by simulating the state-contingent CDX payoffs using the calibrated model parameters and applying the tranche contract terms. Daily model tranche spreads are then computed by valuing the mean simulated state-contingent tranche payoffs using the truncated option-implied state prices. The actual CDX tranche spreads correspond to various tranches referencing the Dow Jones CDX North America Investment Grade index of five-year credit default swaps.

(b) Figure 7: robustness to ruling out extreme market states.

6.8 Figure 6: Downgrade rates and lagged market returns

Read:

- Figure 6 plots the monthly fraction of publicly traded firms with S&P-rated debt that experience downgrades against lagged one-year market returns.
- Downgrade rates are high after poor market performance and low after strong market performance.
- The visual relation supports the model's use of aggregate market returns as an important state variable for credit deterioration.
- The accompanying logistic regression shows that both market returns and firm excess returns help explain downgrade probabilities.

Economic message: Credit quality is not purely firm-specific. Aggregate market conditions predict credit downgrades, so senior tranches exposed to correlated credit deterioration are exposed to systematic economic risk.

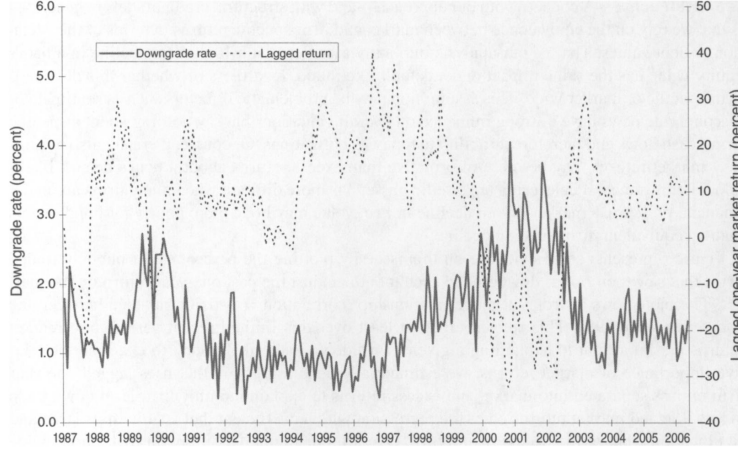


FIGURE 6. US BOND DOWNGRADE RATE AND LAGGED MARKET RETURNS (12/1987–4/2007)

Notes: Downgrades correspond to the fraction of publicly-listed US firms with S&P-rated debt that experienced a downgrade during the calendar month. The lagged one-year return corresponds to the cumulative return on the S&P 500 during the past 12 month period. Using individual firm returns, the corresponding logistic regression is estimated as follows:

$$\text{Prob}(\text{Downgrade}_{i,t} = 1) = \exp(X'\beta) / (1 + \exp(X'\beta)), \text{ where } X'\beta = b_0 + b_1 R_{m,t} + b_2 (R_{i,t} - R_{m,t})$$

The estimated coefficients are displayed in the table below, with standard errors reported in parentheses:

b_0	b_1	b_2	N
3.90	3.27	3.01	129,330
(0.03)	(0.13)	(0.06)	

Figure 6: US bond downgrade rate and lagged market returns.

6.9 Table 6 and Figure 8: Credit risk premia and rating-based pricing

Read:

- Table 6 decomposes spreads into expected-loss compensation and credit risk premia using the credit risk ratio φ .
- Rating-group corporate credit has moderate φ values: AA around 4.3, A around 3.5, BBB around 2.8, and lower-rated credits below that.
- Model-implied φ for senior CDX tranches is much larger. For example, 15–30 has φ of 364.5, and 10–15 has φ of 69.6.
- Figure 8 shows that actual nonequity tranche pricing lies close to the rating-matched credit-market curve, while model-implied tranche pricing lies near the cheapest-to-supply upper bound.
- The equity tranche has actual compensation above the model, consistent with the earlier underpricing result for the first-loss tranche.

Economic message: The table and figure summarize the ratings problem. The market priced tranches as if expected loss were enough, while the model says their disaster-state concentration required much higher compensation.

TABLE 6—CREDIT RISK PREMIA (9/2004–9/2007)

Security	Asset beta	Debt-to-asset ratio	Idiosyncratic asset volatility	Actual spread [bps]	Model spread [bps]	5-yr default probability [bps]	Model recovery rate	φ
<i>Five-year credit default swap indices</i>								
AA	0.85	0.19	0.31	16	16	36	0.41	4.3
A	0.81	0.25	0.29	27	27	73	0.41	3.5
BBB	0.73	0.35	0.27	48	48	162	0.41	2.8
BB	0.44	0.75	0.16	145	145	724	0.43	1.9
B	0.12	1.14	0.05	250	249	1,616	0.47	1.6
<i>Dow Jones CDX North America IG index</i>								
CDX	0.74	0.34	0.27	46	46	153	0.41	2.9
<i>CDX tranches (percentage)</i>								
30–100				NA	1	0.03	0.96	6,106.6
15–30				9	28	4	0.81	364.5
10–15				18	87	22	0.56	69.6
7–10				39	150	77	0.45	25.2
3–7				138	267	599	0.61	7.2
0–3				1,508	914	5,943	0.75	1.7

Notes: This table reports the means of the daily time series of calibrated model parameters and model-implied credit risk premia for various credit market securities. Rating group indices represent the five-year credit default swap spreads implied by spreads of corporate bonds with the associated credit rating. The CDX series is the Dow Jones CDX North America Investment Grade index of five-year credit default swaps. Tranches referencing the CDX are denoted by their lower and upper percentage loss attachment points. The calibrated model parameters are the asset beta (β_A), debt-to-asset ratio (D/A), and idiosyncratic asset volatility (σ_A). The model spread is computed using the calibrated model. The credit risk ratio, φ , is the model spread divided by the loss rate. Loss rate is the annualized spread that is necessary to compensate an investor for expected losses due to default. The model-implied loss rate, five-year default probability and recovery rate are computed under the objective measure, with the auxiliary assumption that the terminal value of the S&P 500 index has a lognormal distribution.

(a) Table 6: credit risk premia.

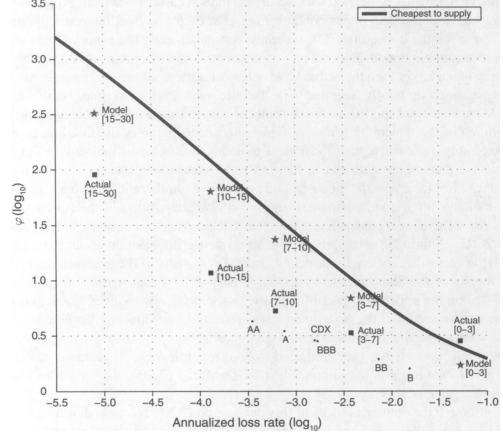


FIGURE 8. CREDIT RISK PREMIA FOR SINGLE-NAME CREDITS AND CDX TRANCHES (9/2004–9/2007)

Notes: This figure displays the time series average of the base 10 log of the credit risk ratio, φ , plotted against the time series average of the base 10 log of the model loss rate for a variety of credit securities. Model (actual) φ is calculated by dividing the model (actual) spread by the model loss rate. Loss rate is the annualized spread that is necessary to compensate an investor for expected losses due to default. φ values for portfolios of single-name corporate credit securities are denoted by circles. The actual CDX tranche φ values are denoted by squares and the model CDX tranche φ values are denoted by stars.

(b) Figure 8: credit risk premia for single-name credit and CDX tranches.

7 Short summary

- The paper argues that senior structured finance tranches are economic catastrophe bonds: they lose money primarily in bad aggregate states.
- Credit ratings summarize default probability or expected loss, but they do not summarize state-contingent exposure.
- The theoretical model shows that pooling and tranching can create the cheapest-to-supply security for a target rating by concentrating default in the worst states.
- The empirical model uses S&P 500 option-implied state prices and a modified Merton-CAPM credit model.
- The model is calibrated to match the CDX index and then used to price CDX tranches.
- Senior and mezzanine CDX tranches have model-implied spreads far above their actual spreads.
- Put-spread approximations in the index-option market deliver similar conclusions and provide an intuitive risk-matched benchmark.
- The overpricing of nonequity tranches is robust to recovery assumptions, constrained volatility extrapolation, and excluding extremely bad market states.
- The equity tranche appears underpriced in the model because it absorbs first losses that are more idiosyncratic and less purely disaster-state risk.
- The central lesson is that structured-finance ratings cannot substitute for asset pricing.

8 Conclusion

8.1 Core conclusion

Coval, Jurek, and Stafford show that securities with the same expected loss can have very different prices because they can lose money in very different economic states. Senior CDO tranches look safe under credit ratings but are risky under state-contingent asset pricing because their losses are concentrated in recessions and market crashes.

8.2 Economic intuition

The intuition is that a dollar is most valuable in bad states. A security that promises to pay except when the economy is very bad is selling insurance against exactly the states where insurance is most valuable. Such a security should pay a large spread. Pooling and tranching make it possible to create securities with low default probability but high exposure to bad-state risk.

8.3 Incremental contribution revisited

The paper contributes by turning a general asset-pricing point into a structured-finance diagnosis. It proves that diversified senior tranches approximate cheapest-to-supply catastrophe claims, then shows empirically that CDX tranches were priced more like rating-matched corporate bonds than like risk-matched market-option positions.

8.4 Implications for investors and regulators

Investors should not treat a AAA or high-grade rating as a complete risk measure. A rating may say that expected loss is small, but it does not say whether the loss arrives in a normal state or a disaster state. Regulators and institutional mandates that rely mechanically on ratings can therefore create demand for securities that appear safe but load heavily on systematic crash risk.

8.5 Limitations

The empirical exercise is a joint test of the model and market efficiency. It relies on a one-factor market-state representation, a representative-firm approximation for the CDX, and extrapolated option-implied state prices. The paper addresses these concerns through robustness checks, but the exact magnitude of mispricing depends on the model. The qualitative message is more robust: expected loss and rating are insufficient for pricing structured finance.

8.6 Takeaway

The enduring takeaway is that the economic state in which default occurs is as important as the probability of default. Senior structured-finance tranches were designed to have low expected losses, but they concentrated those losses in the states where investors most needed payoffs. That is why they are economic catastrophe bonds.